

# Clean Proof of the LAP Variability Bound

Lean formalization note for DGD26

**Purpose.** This note records the paper-facing proof of the finite linear-assignment variability result. The source paper states this result at a high level. The Lean development supplies the detailed finite exchange argument: exact one-for-one insertion, directed alternating splices, a tail-dominance inequality, same-order injectivity for borderline applicants, and the final counting bound.

## Model and theorem

Let  $\mathcal{A}$  be a finite applicant universe and let  $\mathcal{S}$  be a finite set of assignment slots. A feasible assignment for an applicant pool  $X \subseteq \mathcal{A}$  assigns each slot either no applicant or one applicant from  $X$ , and assigns no applicant to two slots. In the capacity-filling regime used here, a feasible assignment fills all slots when  $|X| \geq |\mathcal{S}|$ .

Each applicant-slot pair has an integer weight  $w(a, s)$ . A selector chooses, for each pool  $X$ , a unique global optimum feasible capacity-filling assignment  $A_X$ . The induced choice rule is

$$C(X) = \{a : a \text{ is assigned in } A_X\}.$$

For a fixed slot  $s$ , write  $a <_s b$  when

$$w(a, s) < w(b, s),$$

so  $b$  is strictly preferred to  $a$  at slot  $s$ . Assume every slot has no ties: for distinct applicants  $a, b$ , exactly one of  $a <_s b$  or  $b <_s a$  holds.

Let  $\kappa : \mathcal{S} \rightarrow K$  be any classifier such that

$$\kappa(s) = \kappa(t) \implies (a <_s b \iff a <_t b)$$

for every pair of applicants  $a, b$ . Thus each classifier value is a slot-order class.

**Theorem 1 (linear-assignment variability bound)** *For every pool  $X$ ,*

$$|V_C(X)| \leq |\kappa(\mathcal{S})|,$$

*where  $V_C(X)$  is the set of applicants in  $C(X)$  who can be displaced by adding one fresh applicant. Equivalently, the assignment-induced choice rule has variability bounded by the number of represented slot-order classes.*

## Exact exchange and directed paths

Fix a pool  $X$ , a fresh applicant  $x \notin X$ , and let

$$A = A_X, \quad B = A_{X \cup \{x\}}.$$

Suppose  $y \in C(X) \setminus C(X \cup \{x\})$ . The formalization first proves the exact one-for-one exchange property: under the unique-optimum and capacity-filling hypotheses, the new chosen set is

$$C(X \cup \{x\}) = (C(X) \setminus \{y\}) \cup \{x\}.$$

In particular every old chosen applicant other than  $y$  remains chosen after the insertion.

The old and new assignments define a directed alternating slot graph: there is an edge from slot  $s$  to slot  $t$  if the old occupant of  $s$  in  $A$  is assigned to  $t$  in  $B$ . Let  $s_0$  be the new slot containing  $x$  in  $B$  and let  $s_k$  be the old slot containing  $y$  in  $A$ . The one-for-one exchange yields a directed path

$$s_0, s_1, \dots, s_k$$

from the fresh slot to the lost slot. Write

$$a_i = A(s_i) \quad (0 \leq i \leq k),$$

so  $a_k = y$ , and the path identities are

$$B(s_0) = x, \quad B(s_{i+1}) = a_i \quad (0 \leq i < k).$$

## Two exchange lemmas

**Lemma 1 (dominated old slots lie on the fresh-to-lost path)** *Let  $s$  be an old slot with old occupant  $z = A(s)$ . If*

$$z <_s y,$$

*then  $s$  is on the directed path from  $s_0$  to  $s_k$ .*

*Proof.* Suppose  $s$  is not on that path. Splice the new assignment  $B$  into the old assignment  $A$  on the fresh-root reachable component and keep  $A$  elsewhere. The finite component-closure facts verified in Lean show that the splice is a feasible old-pool assignment, still capacity-filling, and still globally optimal: the complementary splice is feasible for the enlarged pool, and the two objective inequalities cancel.

Because  $s$  is outside the fresh-root component, the spliced old-pool optimum still assigns  $z$  to  $s$ . Because  $y$  is the unique old applicant lost in  $B$ , the same splice still rejects  $y$ . Replacing  $z$  by  $y$  at slot  $s$  would be a feasible one-slot change and would strictly increase the objective, since  $z <_s y$ . This contradicts local optimality, which is derived in Lean from global objective optimality. Therefore  $s$  must lie on the fresh-to-lost path.  $\diamond$

**Lemma 2 (tail dominance along the path)** *For every  $i < k$ ,*

$$w(y, s_i) \leq w(a_i, s_i).$$

*Proof.* Fix  $i < k$ . Define

$$P_i = \sum_{j=i}^{k-1} w(a_j, s_{j+1}), \quad Q_i = \sum_{j=i+1}^k w(a_j, s_j).$$

Construct an old-pool competitor  $E_i$  from  $A$  by rotating the suffix  $s_i, s_{i+1}, \dots, s_k$  forward and putting the lost applicant  $y = a_k$  into  $s_i$ :

$$E_i(s_i) = y, \quad E_i(s_{j+1}) = a_j \quad (i \leq j < k),$$

and  $E_i$  agrees with  $A$  outside the suffix. This is feasible because it only permutes applicants already used by  $A$  on distinct suffix slots. Since  $A$  is old-pool optimal and the outside terms cancel,

$$w(y, s_i) + P_i \leq w(a_i, s_i) + Q_i. \quad (1)$$

Construct an enlarged-pool competitor  $F_i$  from  $B$  by setting the tail  $s_{i+1}, \dots, s_k$  back to the old assignment:

$$F_i(s_j) = a_j \quad (i < j \leq k),$$

and leaving  $B$  unchanged outside that tail. This drops  $a_i$  from the shifted tail and inserts  $y$  at the lost slot, so feasibility follows from the exact one-for-one exchange and no-duplicate assignment facts. Since  $B$  is enlarged-pool optimal and outside terms cancel,

$$Q_i \leq P_i. \quad (2)$$

Combining (1) and (2) gives

$$w(y, s_i) + P_i \leq w(a_i, s_i) + Q_i \leq w(a_i, s_i) + P_i.$$

Canceling  $P_i$  proves  $w(y, s_i) \leq w(a_i, s_i)$ .  $\diamond$

## Same-order injectivity

**Lemma 3 (same-order borderline injectivity)** *Fix  $X$ . Let  $y, z \in V_C(X)$ , and suppose the selected old assignment  $A_X$  assigns  $y$  to slot  $s_y$  and  $z$  to slot  $s_z$ . If  $\kappa(s_y) = \kappa(s_z)$ , then  $y = z$ .*

*Proof.* Assume  $y \neq z$ . By no ties at slot  $s_z$ , either  $z <_{s_z} y$  or  $y <_{s_z} z$ .

If  $z <_{s_z} y$ , use the fresh insertion witnessing  $y \in V_C(X)$ , so  $y$  is the unique old applicant lost. Since  $z$  remains chosen and is below  $y$  at its old slot  $s_z$ , the dominated-slot lemma puts  $s_z$  on the fresh-to-lost path. Applying tail dominance at that path position yields

$$w(y, s_z) \leq w(z, s_z),$$

contradicting  $z <_{s_z} y$ .

If  $y <_{s_z} z$ , the common slot-order hypothesis transports the same strict comparison to  $s_y$ , so  $y <_{s_y} z$ . Now use the fresh insertion witnessing  $z \in V_C(X)$ , with  $z$  as the unique old applicant lost. The same argument puts  $s_y$  on the corresponding fresh-to-lost path and gives

$$w(z, s_y) \leq w(y, s_y),$$

contradicting  $y <_{s_y} z$ .

Both cases are impossible, so  $y = z$ .  $\diamond$

## Counting

For each borderline applicant  $y \in V_C(X)$ , let  $s_y$  be the unique slot where  $A_X$  assigns  $y$ . Define

$$\phi(y) = \kappa(s_y).$$

The map is well-defined because  $y \in C(X)$  and assignments do not duplicate applicants. The same-order injectivity lemma says exactly that  $\phi$  is injective. Since both sets are finite,

$$|V_C(X)| \leq |\kappa(\mathcal{S})|.$$

This proves the linear-assignment variability bound.